Experiment No: 01

AIM: Git commands to create a local repository, update it, and push it to the Git Hub.

SOURCE:

Step1:

- Move to the Drive (D)

```
badbo@GADDAMVENU MINGW64 ~
$ cd d:
```

- Create a New Directory
  This command creates a folder where your project will live.

```
badbo@GADDAMVENU MINGW64 /d
$ mkdir VENU
```

- Move to directory, VENU.

```
badbo@GADDAMVENU MINGW64 /d
$ cd VENU
```

- Create a local repository on the New Directory (VENU)

```
badbo@GADDAMVENU MINGW64 /d/VENU
$ mkdir local_repos
```

- Move to local_repos:

```
badbo@GADDAMVENU MINGW64 /d/VENU
$ cd local_repos
```

Step2:

- Initialize Git:
  git init initializes a new Git repository locally, allowing Git to start tracking changes.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos
$ git init
Initialized empty Git repository in D:/VENU/local_repos/.git/

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ |
```

- Configure Git with Username and Email

Git needs your identity to record who makes commits. This sets your username and email globally (once for all repositories on your machine).

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git config --global user.name "VenuKrishna@1204"

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git config --global user.email "gaddamvenu12042004@gmail.com"
```

Step 3:

- Create a New File in the Repository
  This command creates file1.txt and writes initial content into it. You can also use editors like notepad or touch depending on your OS.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat >>file1.txt
VENU
FAROOK
SIMHADRI
ARJUN
BABU
HARSHA

[1]+  Stopped                 cat >> file1.txt
```

- Display the results of file1.txt:

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat file1.txt
VENU
FAROOK
SIMHADRI
ARJUN
BABU
HARSHA
```

- Check the Status of the Repository
  Displays the state of the working directory and staging area. Shows which files are untracked, modified, or staged for commit.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt

nothing added to commit but untracked files present (use "git add" to track)
```

- Create the multiple files in the local repository.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat >>f2.txt
VENU

[2]+  Stopped                 cat >> f2.txt

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat >>f3.txt
SIMHADRI

[3]+  Stopped                 cat >> f3.txt

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat >>f4.docx
HARSHA VARDHAN

[4]+  Stopped                 cat >> f4.docx
```

- Display multiple files

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat f2.txt
VENU

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat f3.txt
SIMHADRI

badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ cat f4.docx
HARSHA VARDHAN
```

Step4:

- Check the Status of the Repository
  Displays the state of the working directory and staging area. Shows which files are untracked, modified, or staged for commit.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        f2.txt
        f3.txt
        f4.docx
        file1.txt

nothing added to commit but untracked files present (use "git add" to track)
```

- Add the File to the Staging Area

  Moves the file from the working directory to the staging area. This tells Git that you want to include this file in the next commit.

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git add .
warning: in the working copy of 'f2.txt', LF will be replaced by CRLF the next time
warning: in the working copy of 'f3.txt', LF will be replaced by CRLF the next time
warning: in the working copy of 'f4.docx', LF will be replaced by CRLF the next time
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next ti
```

- Commit the File

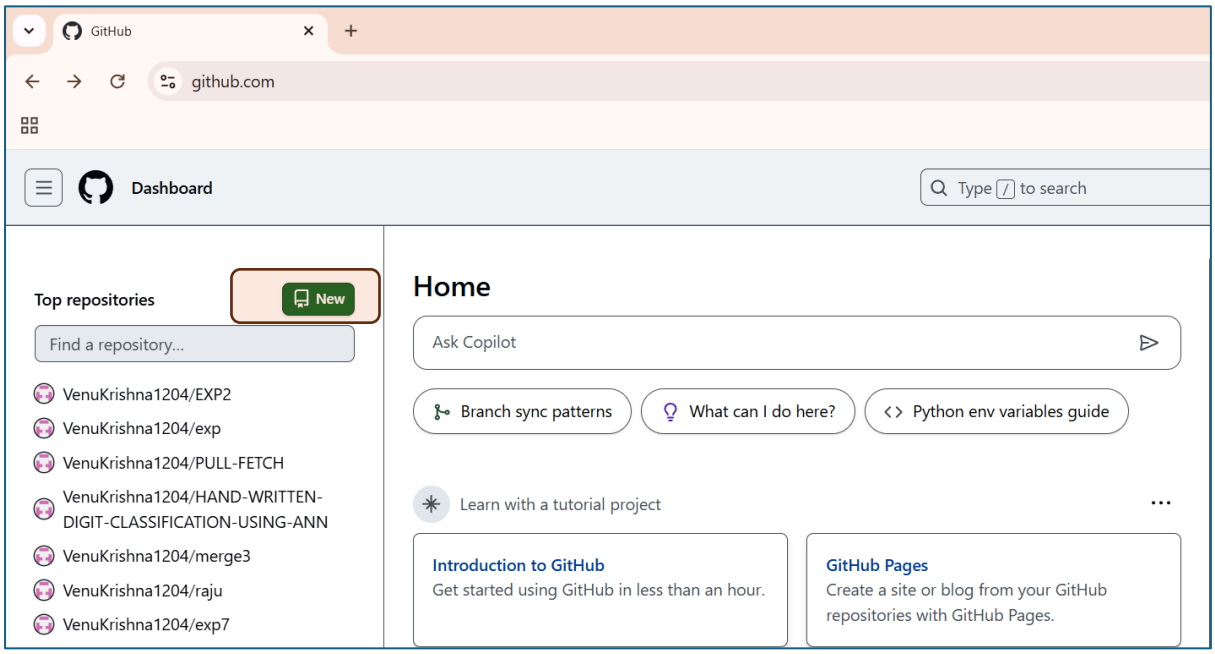
  Records a snapshot of the staging area to the Git repository. The -m flag lets you add a message describing the change.

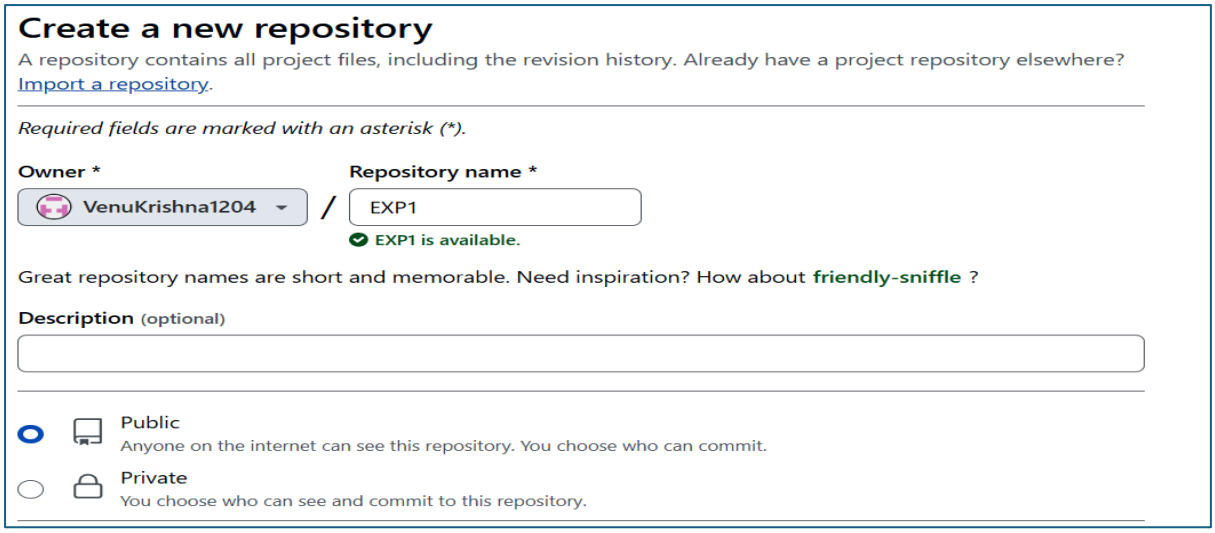
```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git commit -m "commited"
[master (root-commit) a82e32b] commited
 4 files changed, 9 insertions(+)
 create mode 100644 f2.txt
 create mode 100644 f3.txt
 create mode 100644 f4.docx
 create mode 100644 file1.txt
```

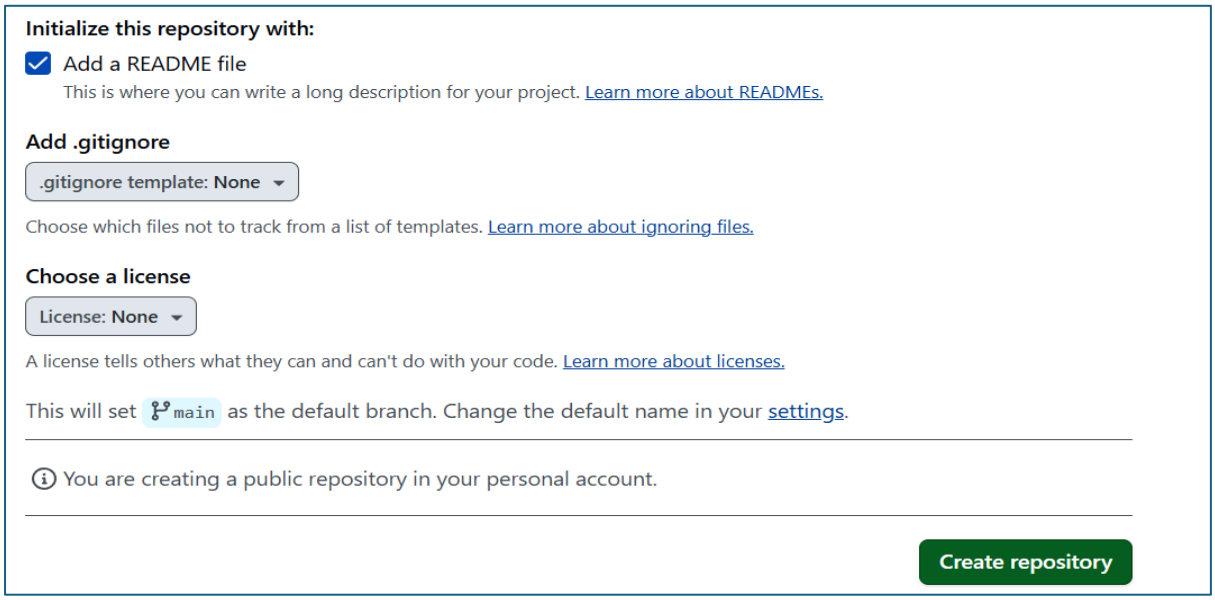
Step5:

- **Create a Remote Repository on GitHub**

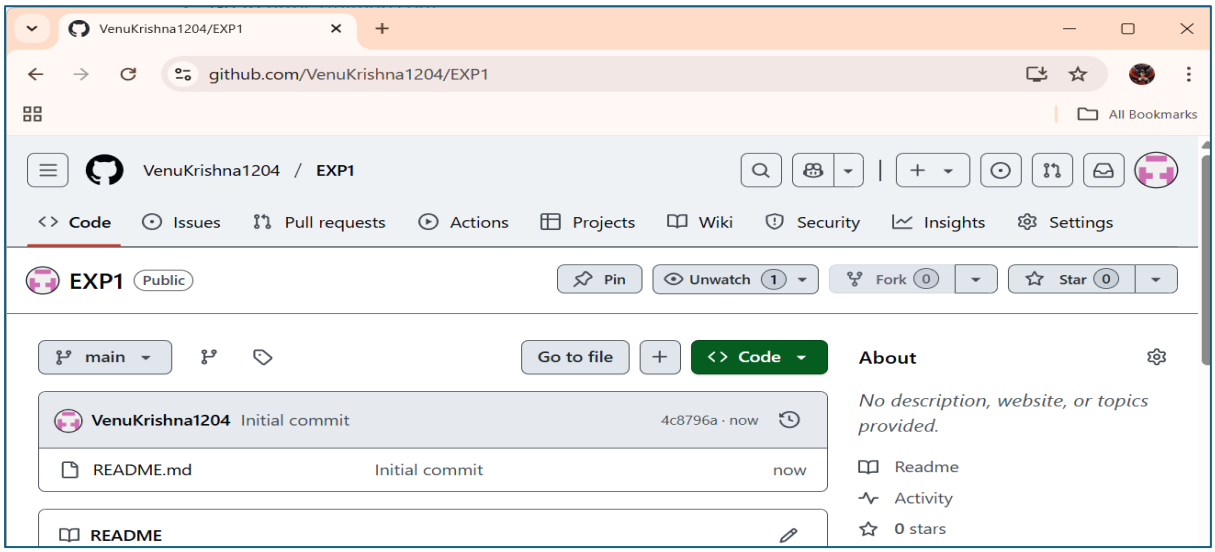
  Go to https://github.com
  Click **New Repository**
  Enter repository name (e.g., my-repo)
  Leave "Initialize this repository with a README" unchecked
  Click **Create Repository**







❖ **Create a repository**



Step:06

- **Link Local Repository to GitHub Remote**

Links your local Git repo to a remote GitHub repo. Replace the URL with your actual GitHub repository link.(https://github.com/VenuKrishna1204/EXP1.git)

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git remote add origin https://github.com/VenuKrishna1204/EXP1.git
```

- Check the git log status

```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git log
commit a82e32b9fe5ee641f3c90798267e30aa71368663 (HEAD -> master)
Author: VenuKrishna@1204 <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 15:12:03 2025 +0530

    commited
```

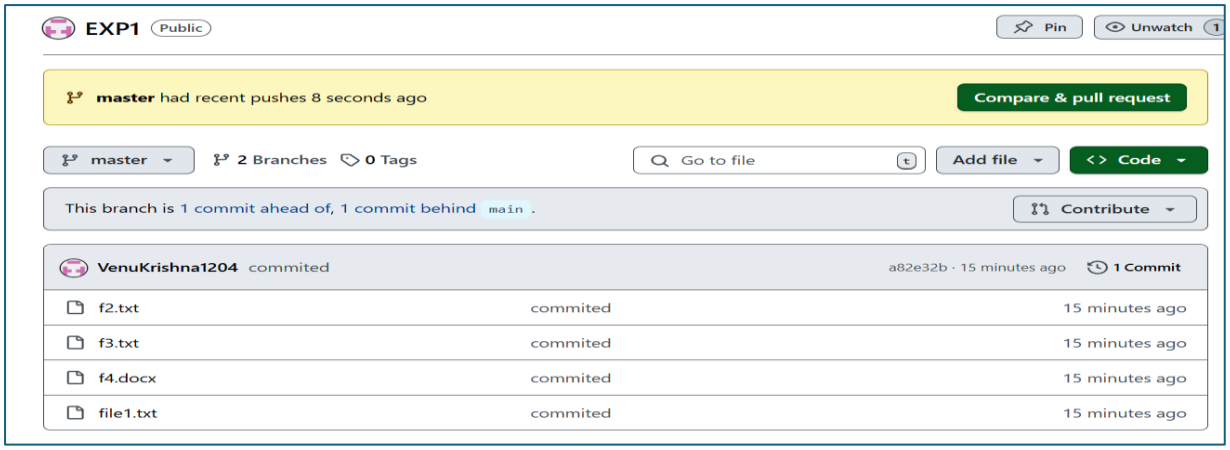
- **Push Files to GitHub**

Pushes your local commits to the GitHub server. The -u flag sets the upstream, so future pushes can use git push alone.

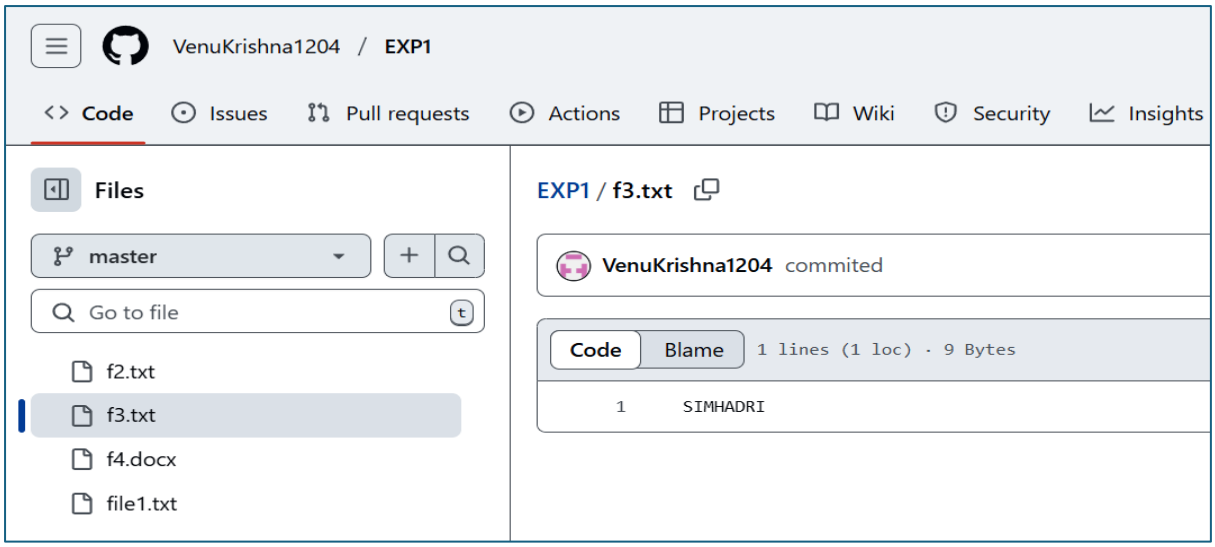
```
badbo@GADDAMVENU MINGW64 /d/VENU/local_repos (master)
$ git push origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 399 bytes | 199.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/VenuKrishna1204/EXP1/pull/new/master
remote:
To https://github.com/VenuKrishna1204/EXP1.git
 * [new branch]      master -> master
```

Step:07

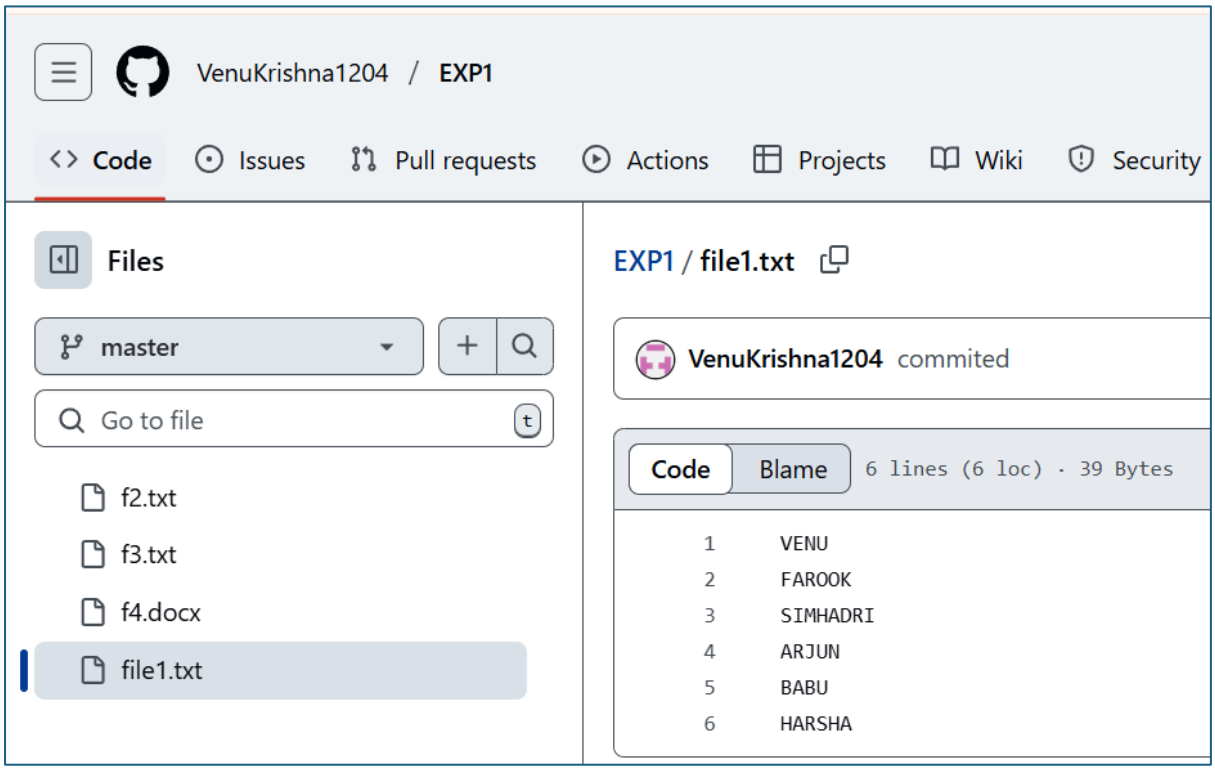
- **Verify Files on GitHub:** Go to your GitHub repository URL in a browser. You will see the files (file1.txt, file2.txt) you committed and pushed.



❖ Verify the details of the f3.txt



❖ Verify the details of the file1.txt



Experiment :02

AIM: Git commands are used to demonstrate cloning, updating, and pushing the changes to the remote repository.

Source code:

Step1:

- Move to the Drive (D)

```
badbo@GADDAMVENU MINGW64 ~
$ cd d:
```

- Create a New Directory
  This command creates a folder where your project will live.

```
badbo@GADDAMVENU MINGW64 /d
$ mkdir VENU
```

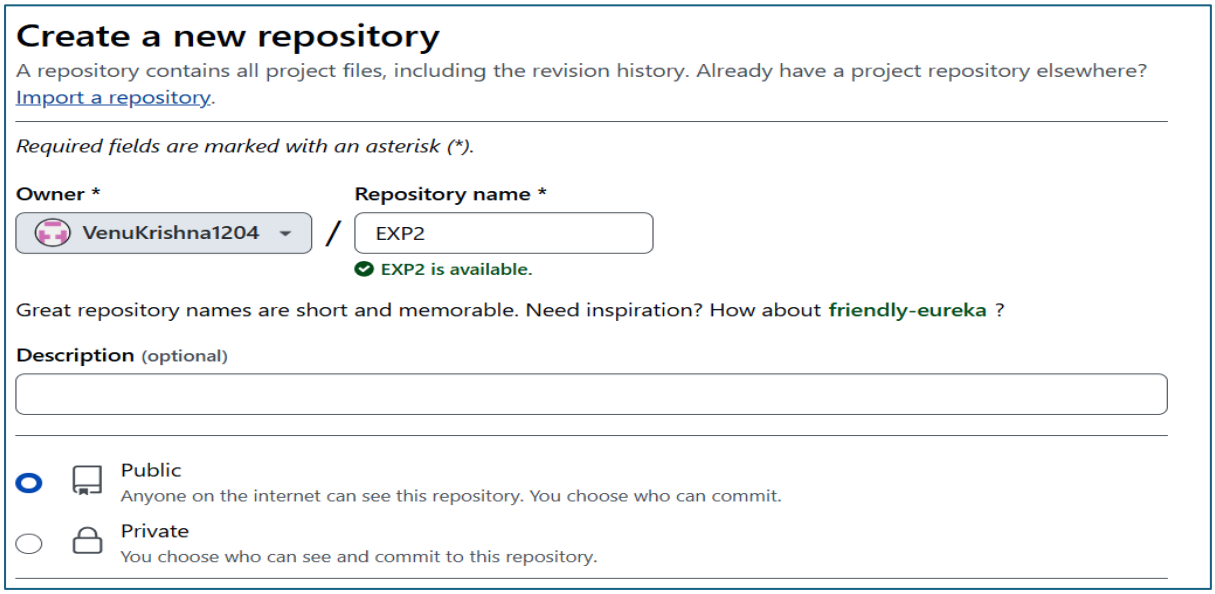
- Move to directory, VENU.

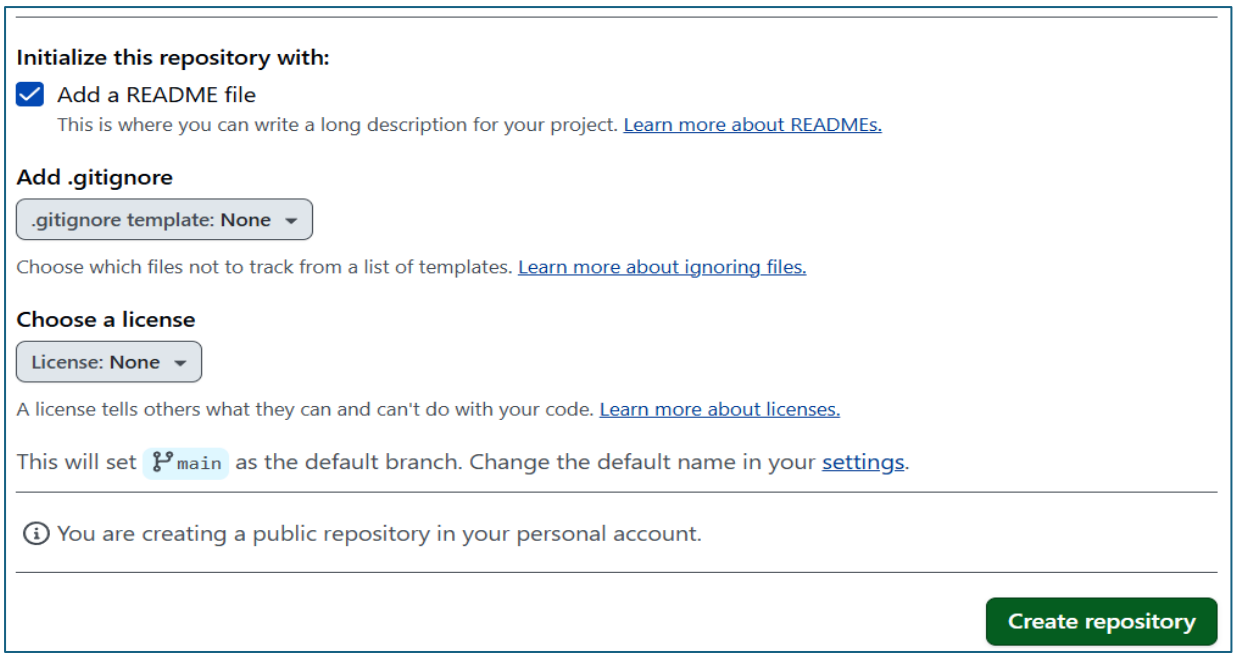
```
badbo@GADDAMVENU MINGW64 /d
$ cd VENU
```

Step2:

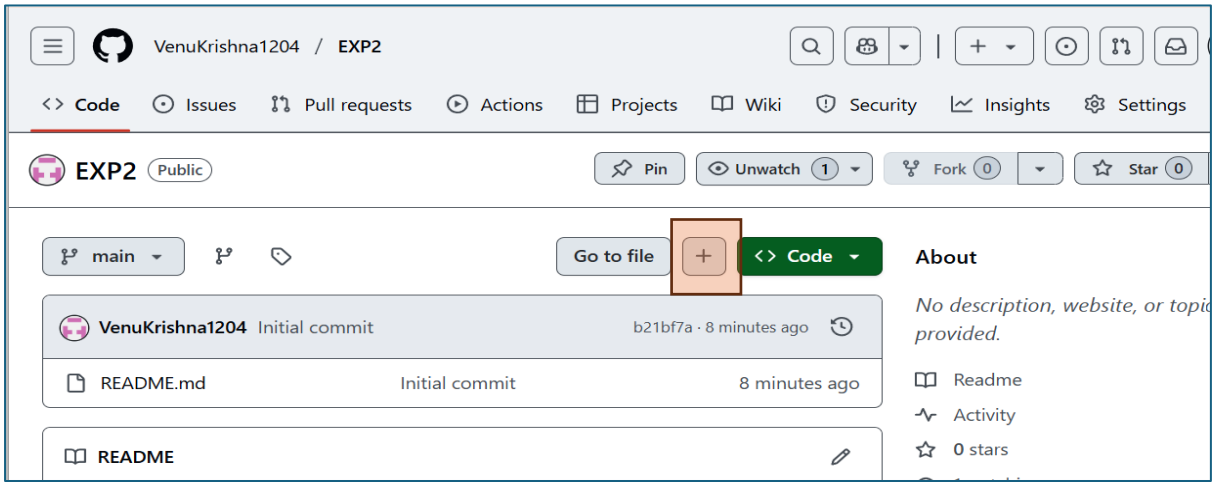
- **Use or Create a Repository on GitHub**
  To create a new one:

  1. Visit https://github.com
  2. Click on **New repository**. Provide repository name and click **Create Repository**

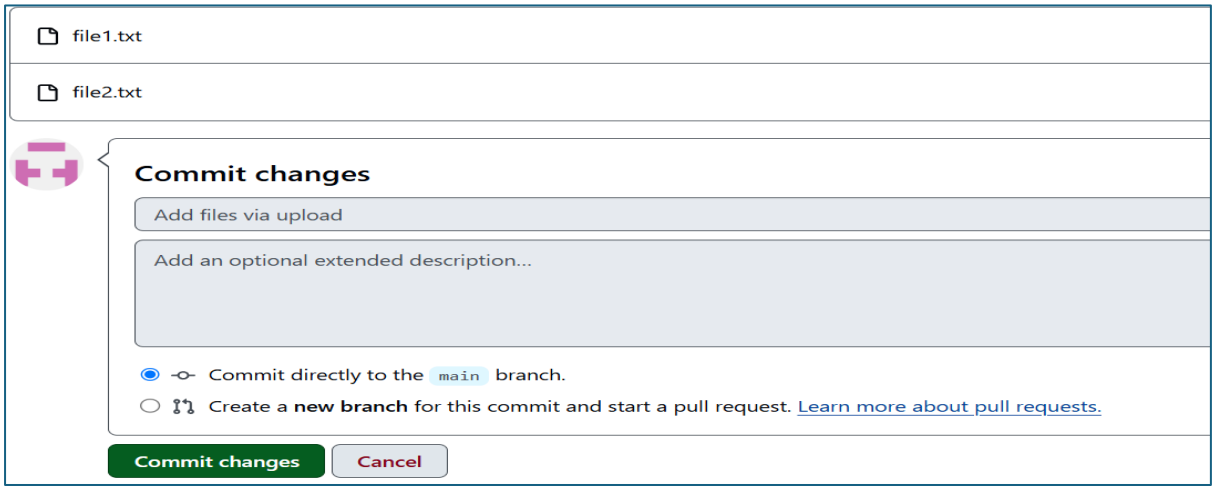


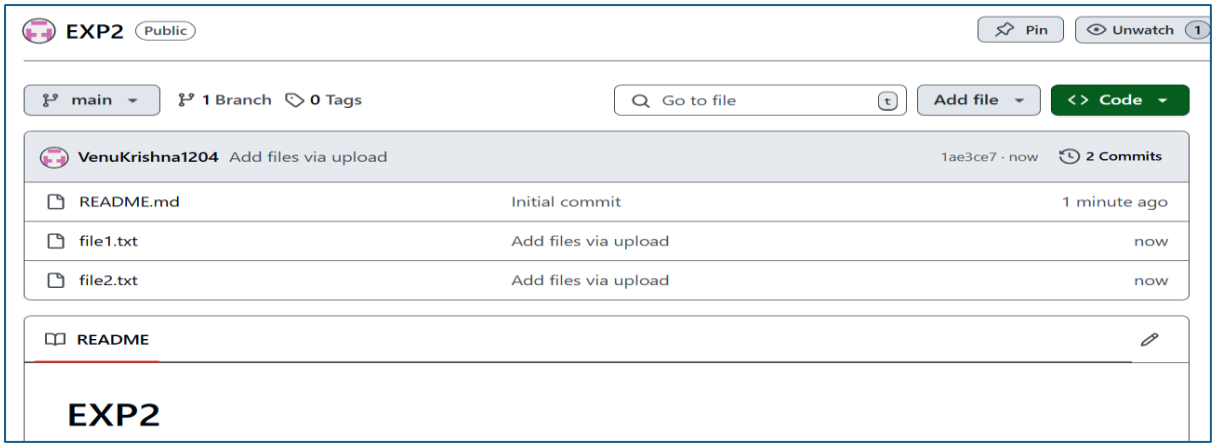


❖ Add files to the repository.



❖ Click the Plus symbol and add files to the repository
Files are **file1.txt, file2.txt**





Step3:

- **Clone the Remote Repository into Local System**
  This command downloads the entire contents of the remote GitHub repository to your local system. (https://github.com/VenuKrishna1204/EXP2.git)

```
badbo@GADDAMVENU MINGW64 /d/venu
$ git clone https://github.com/VenuKrishna1204/EXP2.git
Cloning into 'EXP2'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.
```

- Check the files using ls command. And **Navigate into the Cloned Repository**

```
badbo@GADDAMVENU MINGW64 /d/venu
$ cd EXP2/

badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$
```

- Verify the files

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ ls
README.md  file1.txt  file2.txt
```

Step 4:

- **Modify a File Locally and print file.**
  Open or edit an existing file in the repository to make some changes.

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ cat >>file1.txt
Banana
kiwi
Hi
Hello

[1]+  Stopped                 cat >> file1.txt

badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ cat file1.txt
Banana
kiwi
Hi
Hello
```

- **Check Status and Add Changes to Staging Area**
  Check what changes were made and stage the modified file for commit.

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

- Add the file1.txt to staging Area.

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF
 touches it
```

- Verify the status

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file1.txt
```

- **Commit the Changes**
  Create a new commit that includes the staged changes and provide a commit message.

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git commit -m "commited"
[main 6960368] commited
 1 file changed, 4 insertions(+)
```

- Verify the Status

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

- Verify the log files

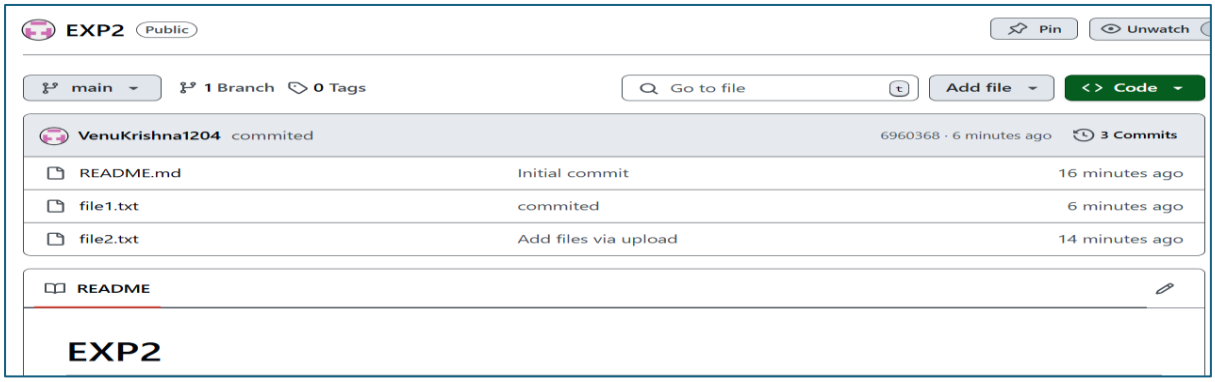
```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git log
commit 696036839a033191047b697d7d80121e00012296 (HEAD -> main)
Author: VenuKrishna@1204 <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 19:30:11 2025 +0530

    commited

commit 1ae3ce78493dc47d19101a4b94cba101b45d5312 (origin/main, origin/HEAD)
Author: GADDAM VENU <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 19:22:43 2025 +0530

    Add files via upload

commit ffd7c72d909a3c7d3064ebb760b9c9b3b89ed0d5
Author: GADDAM VENU <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 19:21:09 2025 +0530

    Initial commit
```

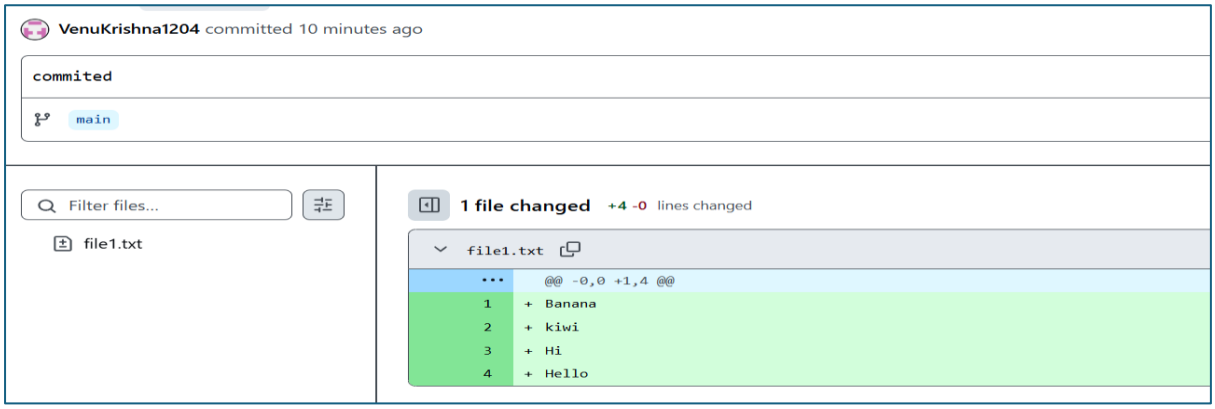
- **Push Changes to GitHub**
  Upload your local commits to the remote repository on GitHub.

```
badbo@GADDAMVENU MINGW64 /d/venu/EXP2 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 328 bytes | 328.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/VenuKrishna1204/EXP2.git
   1ae3ce7..6960368  main -> main
```

Step: 05

- **Verify Files on GitHub:** Go to your GitHub repository URL in a browser. You will see the files (file1.txt, file2.txt) you committed and pushed.





Experiment -03

AIM: Using the fetch and git merge commands, performing changes and pushing changes into the repository.

Source code:

Step1:

- I have already exiting the existing remote repository.
- Creating a repository and change your directory to that repository and change your directory, for this we use the "mkdir" command.
- Move to the Drive (D)

```
badbo@GADDAMVENU MINGW64 ~
$ cd d:
```

- Create a New Directory
  This command creates a folder where your project will live.

```
badbo@GADDAMVENU MINGW64 /d
$ mkdir VENU
```

- Move to directory, VENU.

```
badbo@GADDAMVENU MINGW64 /d
$ cd VENU
```

- Create a local repository on the New Directory (VENU)

```
badbo@GADDAMVENU MINGW64 /d/venu
$ mkdir pull_fetch
```

- Move to pull_fetch:

```
badbo@GADDAMVENU MINGW64 /d/venu
$ cd pull_fetch
```

Step2:

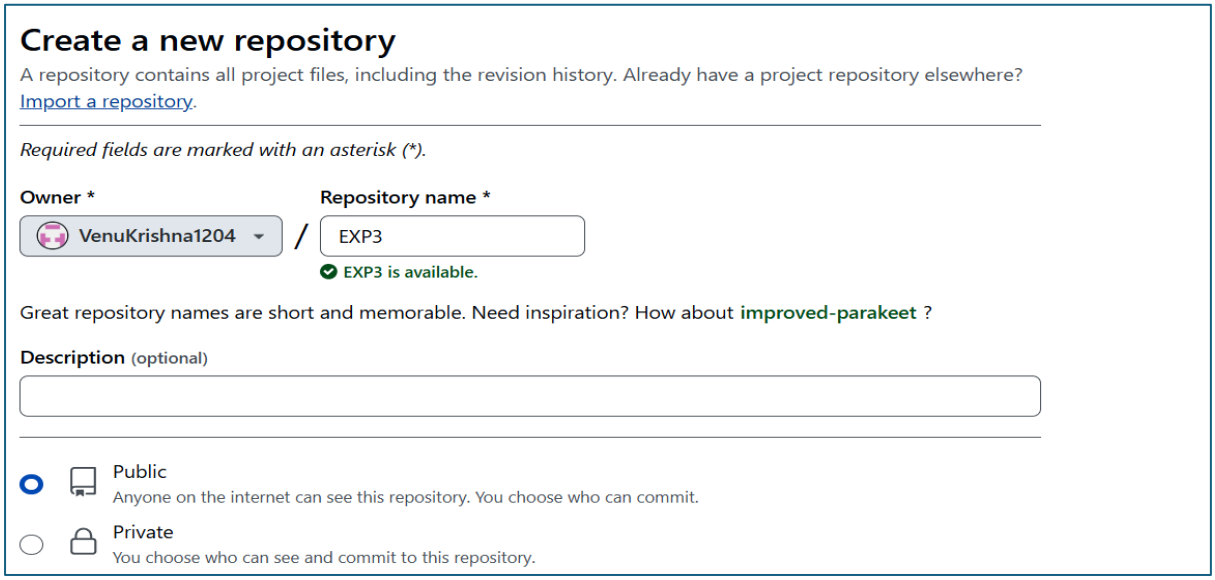
- Initialize Git:
  git init initializes a new Git repository locally, allowing Git to start tracking changes.

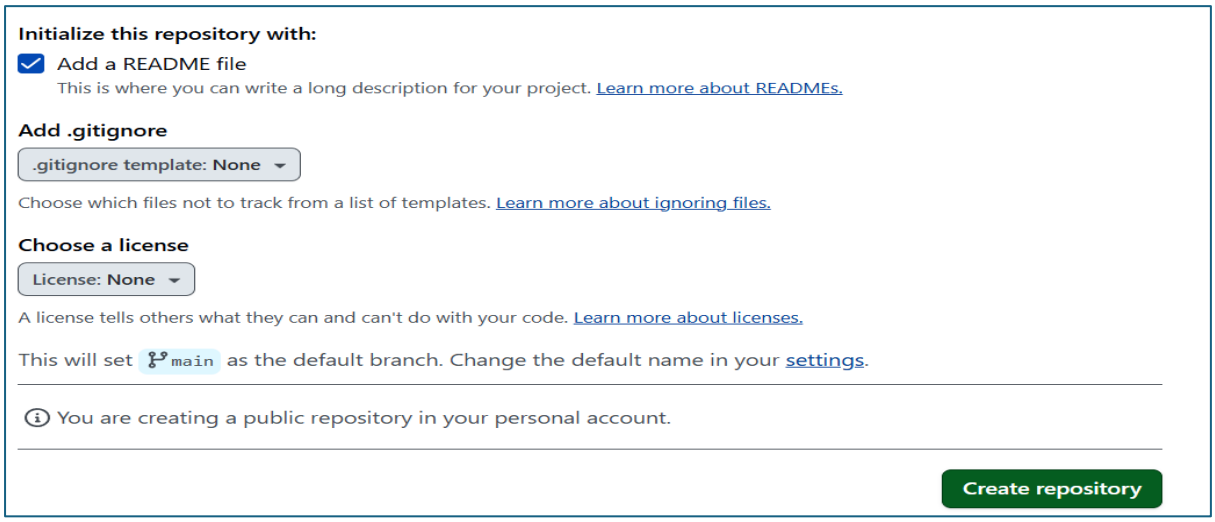
```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch
$ git init
Initialized empty Git repository in D:/VENU/pull_fetch/.git/
```

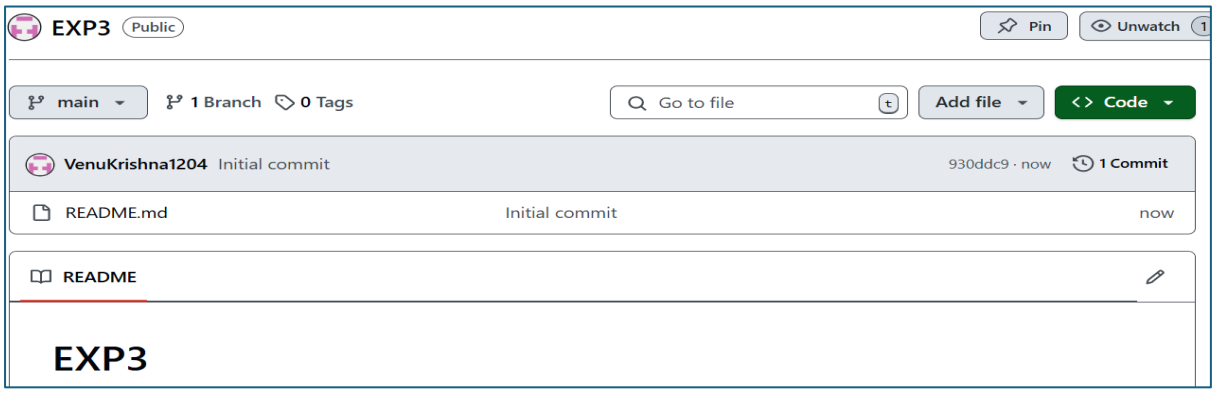
Step :03

- **Use or Create a Repository on GitHub**
  To create a new one:

  3. Visit https://github.com
  4. Click on **New repository**. Provide repository name and click **Create Repository**







STEP: 03

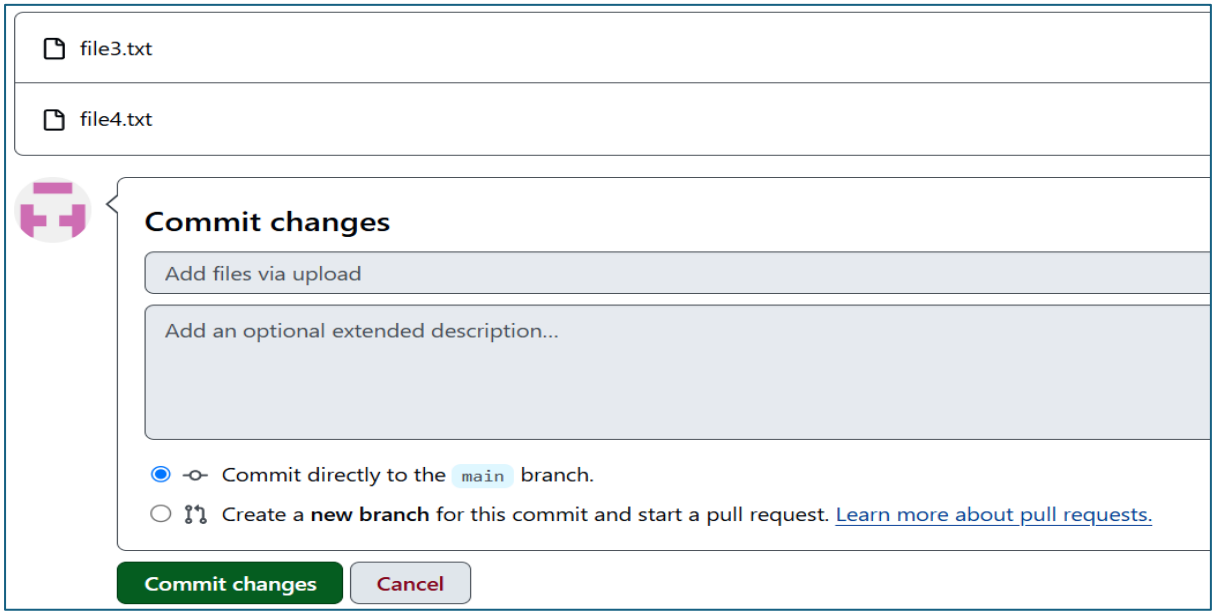
- Use the git remote command by adding the remote repository.

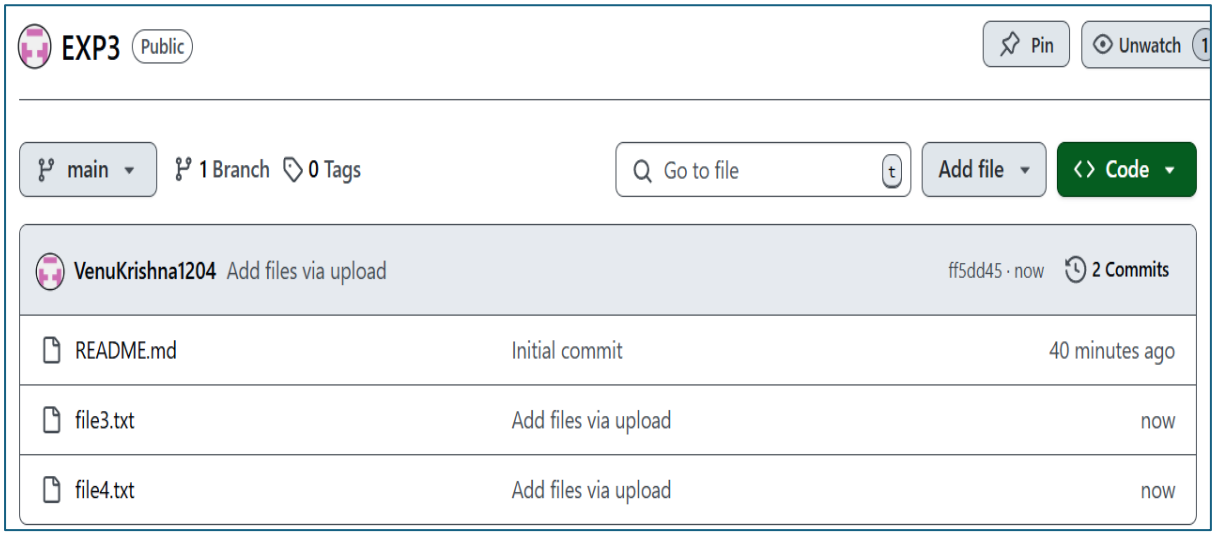
```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git remote add origin https://github.com/VenuKrishna1204/EXP3.git
```

- Check the status

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

- Add the files to remote repository





❖ Pull the repository from the main using the git pull command.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 982 bytes | 35.00 KiB/s, done.
From https://github.com/VenuKrishna1204/EXP3
 * branch            main       -> FETCH_HEAD
   930ddc9..ff5dd45  main       -> origin/main
Updating 930ddc9..ff5dd45
Fast-forward
 file3.txt | 3 +++
 file4.txt | 1 +
 2 files changed, 4 insertions(+)
 create mode 100644 file3.txt
 create mode 100644 file4.txt
```

❖ View all the files in the remote repository using ls command.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ ls
README.md  file3.txt  file4.txt
```

❖ Modify any one of the files by adding few lines to the using **cat** command.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ cat file3.txt
HI
HELLO
WELCOME
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ cat >>file3.txt
Complete DevOps record
Welcome to AIML

[1]+  Stopped                 cat >> file3.txt

badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ cat file3.txt
HI
HELLO
WELCOMEComplete DevOps record
Welcome to AIML
```

❖ Verify the status

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file3.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

❖ Add the File to the Staging Area

Moves the file from the working directory to the staging area. This tells Git that you want to include this file in the next commit.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git add file3.txt
```

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file3.txt
```

- Commit the File
  Records a snapshot of the staging area to the Git repository. The -m flag lets you add a message describing the change.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git commit -m "commited"
[master 96707fd] commited
 1 file changed, 2 insertions(+), 1 deletion(-)
```

- Verify the Status.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git status
On branch master
nothing to commit, working tree clean
```

- Verify the git log.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git log
commit 96707fd29ae9295813a6c6432eb5cbe1fb2e38b5 (HEAD -> master)
Author: VenuKrishna@1204 <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 21:00:00 2025 +0530

    commited

commit ff5dd4564ab628943c96a560933870fc14b19a1b (origin/main)
Author: GADDAM VENU <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 20:47:38 2025 +0530

    Add files via upload

commit 930ddc96783f36bc6e7217863199119f203d12a6
Author: GADDAM VENU <gaddamvenu12042004@gmail.com>
Date:   Sat May 10 20:07:25 2025 +0530

    Initial commit
```

Step :04

- Use The git fetch command is used to **download commits, files, and references from a remote repository** into your local repository.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git fetch origin main
From https://github.com/VenuKrishna1204/EXP3
 * branch            main       -> FETCH_HEAD
```

- Verify the ls command

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ ls
README.md  file3.txt  file4.txt
```

- View the file3.txt

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ cat file3.txt
HI
HELLO
WELCOMEComplete DevOps record
Welcome to AIML
```

- Use the checkout main command to know the files status whether they are ready to merge or not and have. They already committed or not.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (master)
$ git checkout main
branch 'main' set up to track 'origin/main'.
Switched to a new branch 'main'
```

- Verify the ls command

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ ls
README.md  file3.txt  file4.txt
```

- View the contents in the modified file.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ cat file3.txt
HI
HELLO
WELCOME
```

- Merge the local repository with the modified content in the files to the remote repository.

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ git merge master
Updating ec23474..b62fc4f
Fast-forward
 file3.txt | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
```

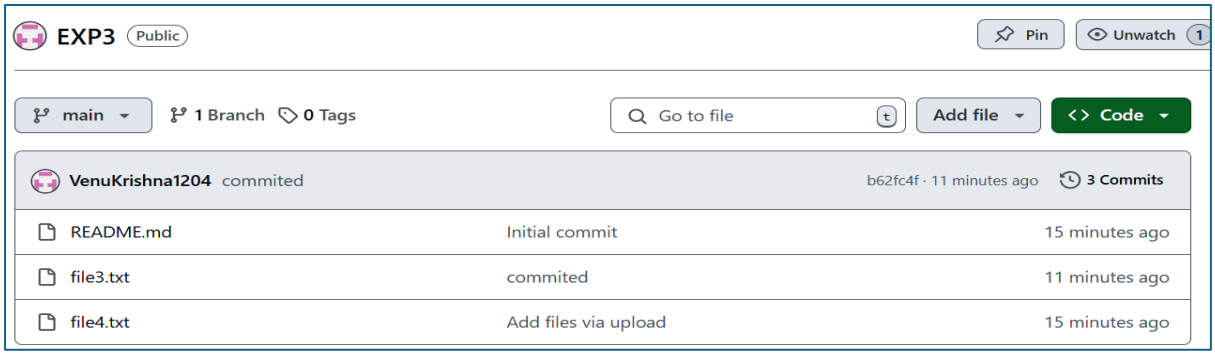
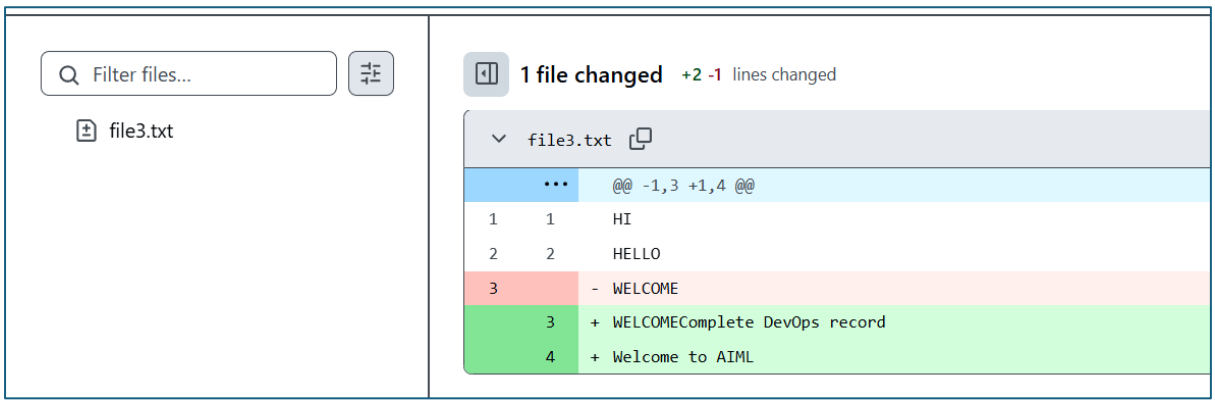
- Verify the ls command

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ ls
README.md  file3.txt  file4.txt
```

- View the details in the file3.txt

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ cat file3.txt
HI
HELLO
WELCOMEComplete DevOps record
Welcome to AIML
```

- Push the files into remote repository

```
badbo@GADDAMVENU MINGW64 /d/venu/pull_fetch (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 366 bytes | 366.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/VenuKrishna1204/EXP3.git
   ec23474..b62fc4f  main -> main
```

- Verify the files in remote repository and view the contents(file3.txt) in remote repository